

MARIUS EMANUEL HEES

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Erasmus School of Economics, Erasmus University Rotterdam
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Up to date: August 2026

PLACEMENT: Tinbergen Institute

Prof. Eric Bartelsman, placement director
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Christina Månsson, placement assistant
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MAIN INTERESTS

I am an economist working on urban, spatial, environmental, and development economics. My research studies how space and networks shape economic activity, and how these dynamics, through migration, trade, infrastructure, and climate adaptation, reshape economic outcomes. Combining structural modeling, field experiments, and spatial methods with satellite and micro data, I aim to address key challenges in climate risk and resilience, migration and urbanization, urban labor markets and informality, transport and infrastructure, trade networks, and structural transformation. I am on the 2026–2027 job market.

FIELDS OF CONCENTRATION

PRIMARY

Development Economics
Urban, Spatial and Trade Economics
Environmental and Climate Economics

SECONDARY

Applied Econometrics
Migration and Labor Economics
Quantitative Spatial Economics

DISSERTATION TITLE

“Climate, Migration and Structural Transformation: Satellite Data, Field Experiments and General Equilibrium Models for Resilient Development”

Expected Date of Completion: Spring 2027

Principal Advisors: Professor Frank van Oort and Dr. Michiel Gerritse

PAPERS

JOB MARKET PAPER

“Segmented Labor Markets and Structural Transformation”

Abstract: Urbanization without formalization is a major barrier to structural transformation and sustained growth in many developing economies. As workers move to cities, many cannot access productive formal employment and remain stuck in low-productivity informal work, reducing the gains from urban migration. This paper quantifies how formalization shapes growth, welfare and distribution. I develop a dynamic spatial equilibrium model with segmented labor markets in which workers move between rural agriculture, urban informal employment and urban formal employment, and access to formal work is constrained by the capacity of firms and institutions to absorb informal workers. When this capacity is weak, migration enlarges the informal labor pool, slows formalization, and can generate a self-reinforcing informality trap. Quantifying the model for Ethiopia shows that limited access to formal employment constrains structural transformation: raising formalization capacity increases long-run GDP by 13.1 percent, national welfare by 9.7 percent, and the urban formal employment share from 41.1 to 67.5 percent. Counterfactuals show that access to formal employment shapes policy effectiveness and distribution, generating sharply different effects across workers within the same city that single-labor-market models miss. The framework applies whenever access to a location does not imply access to all of its labor markets.

Presented at: North American Meeting of the Urban Economics Association (Chicago, 2026); Dutch Urban Economics and Trade Workshop (2025, 2026).

WORKING PAPERS

“(Mis)measuring Local Weather Shocks: Evidence from Drought and Migration in Ethiopia”

(with Michiel Gerritse and Frank van Oort). Under review.

Abstract: Extreme weather shocks strike locally, yet their effects are typically estimated from exposure aggregated across districts, regions or countries. We show that this aggregation systematically biases impact estimates and propose a simple correction. Using high-resolution rainfall and satellite-derived population data for more than 13,000 Ethiopian settlements (2000–2016), we find that drought reduces population in vulnerable settlements by about 2.5 percent, concentrated among working-age adults and coinciding with falling food production, income and health. Measured at progressively coarser scales, the impact disappears, even at resolutions common in the literature. We decompose the bias into three sources (partial exposure misclassified, local vulnerability averaged away, area weights misrepresenting where people live) and derive a fix: measure each shock at fine scale, weight by local vulnerability, then aggregate. The correction recovers estimates in areas roughly four times larger than those at which the standard binary indicator fails; simulations confirm its performance. Building exposure locally before aggregating is a low-cost change that can substantially improve empirical practice.

Presented at: 6th GCD Workshop on Regional Development and Innovation (A Coruña, 2024); Regional Science Workshop on Remote-Sensed Identification and Networks (Oviedo, 2024); PSE Summer School on Migration (2023); ESE/TI internal seminars.

“Memory, Expectations, and Climate Adaptation”

(with Frank van Oort). Draft, September 2026. Pre-registered field experiment (AEA RCT Registry, 2026).

Abstract: Smallholder farmers face increasingly frequent weather shocks, yet take-up of insurance and other protection remains low. We show that part of this demand problem lies in how farmers form beliefs about future weather. In a survey experiment with 686 farmers in eastern Uganda, we elicit memories of past seasons and expectations for the coming one, benchmark them against remotely sensed weather records, and experimentally provide information on local weather history. Memory is strongly compressed—large differences in the record become small differences in recall—and this compression carries into expectations, while memorable extreme events form a separate, less compressed channel. Beliefs matter: farmers expecting adverse weather plan more adaptation. A simple visual summary of local weather history moves expectations toward the record, most for those furthest away; literacy predicts accuracy but not responsiveness. Belief formation is thus a demand-side margin in climate adaptation, and accessible information a low-cost complement to protection policies.

Presented at: ESE/TI internal seminars.

EDUCATION

	LEVEL	DATE	FIELD
Erasmus University Rotterdam & Tinbergen Institute	PhD	2027 (exp.)	Economics
University of Chicago (Host: E. Rossi-Hansberg)	Visiting PhD	2024	Economics
University of Oxford	MPhil	2022	Economics
Erasmus School of Economics	BSc	2019	Economics
Harvard University	Visiting BSc	2018	Economics

PROFESSIONAL EXPERIENCE

Research Assistant to Dr. Niclas Moneke, University of Oxford	2021
Spatial development project on mobile-phone network expansion and economic development	
Research Assistant, ifo Institute for Economic Research, Munich	2020
Project with Prof. Niklas Potrafke on trade networks (Center for Public Finance and Political Economy)	
Research Assistant and Consultant, United Nations ESCAP, Bangkok	2019
Cross-national rail connectivity across the Asian continent (Transport Connectivity and Logistics Section)	
Research Assistant, University of Tehran	2019
Data collection on local labor market dynamics	

TEACHING

Lecturer, Urban Economics (MSc), Erasmus School of Economics 2025
 Foundational course of the MSc programme in Urban, Port and Transport Economics. Designed and taught the module on developing-country cities: lecture, R-based problem set and seminar on general-equilibrium analysis of urban transport policy, and assessment

Lecturer, Firm Location Choice and Productivity in Cities (BSc), Erasmus School of Economics 2022–2025
 BSc Strategy Economics seminar, four cohorts: weekly session reports, literature surveys, student paper presentations and quizzes; grading and feedback for 100–120 students per cohort

Thesis Supervision, Erasmus School of Economics 2022–2026
 BSc theses in Strategy Economics (15) and Urban, Port and Transport Economics (3); MSc theses in Strategy Economics (19) and Urban, Port and Transport Economics (4). Topics include optimal transport-network allocation, carbon pricing and local pollution, energy-crisis resilience of firms, and urban regeneration.

SKILLS

Programming	R, Julia, Python, Stata, MATLAB, Bash, Git, $\LaTeX$
Spatial and remote sensing	ArcGIS, QGIS, Google Earth Engine; large-scale satellite-data pipelines
High-performance computing	Snellius national supercomputer (SURF); parallel and batch workflows
Languages	German (native), English (fluent), French (C1), Dutch (A2)

WORKSHOPS, SEMINARS AND CONFERENCES

North American Meeting of the Urban Economics Association, Chicago	2026
Dutch Urban Economics and Trade Workshop, Groningen	2026
Dutch Urban Economics and Trade Workshop, Rotterdam	2025
6th GCD Workshop on Regional Development and Innovation, A Coruña	2024
Regional Science Workshop on Remote-Sensed Identification and Networks, Oviedo	2024
Summer School on Migration (Giovanni Peri), Paris School of Economics	2023
ESE and Tinbergen Institute internal seminars	2023–2026

FELLOWSHIPS, GRANTS AND AWARDS

Erasmus Research Institute of Management (ERIM)	€11,000	2025–2026
Field Experiment in Uganda: PhD Support and Data Collection Grants		
Tinbergen Institute, ESE and Erasmus Professor funds	€16,000	2025–2026
Field Experiment in Uganda: Internal research funding		
Erasmus Trustfonds	€5,000	2024
Research visit, University of Chicago		
German Academic Scholarship Foundation (Studienstiftung)	€40,000	2020–2022
MPhil scholarship		
German Academic Scholarship Foundation (Studienstiftung)	€45,000	2016–2019
BSc scholarship		

OTHER INFORMATION

Citizenship: Germany and France (EU); United Kingdom pre-settled status

REFERENCES AND DISSERTATION COMMITTEE

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Prof. Esteban Rossi-Hansberg
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